

Project Guidelines

The project revolves around the analysis of experimental data from the MICrONS dataset, the largest imaging of a mammalian brain. During the experiments, the mouse watch patterns and movies. The data collected includes anatomical connectivity of neurons, as well neural activity across the brain. *Your task is to select a project, ask a related question and try to answer it through the analysis of the dataset.*

You will work in groups of 5 and deliver an oral presentation to the class. You must also deliver a 2-page written report, as well as the code you used to perform the analysis.

Selecting a project

We propose a list with several potential projects. You are also welcome to suggest your own idea, as long as it is feasible.

The questions formulated for each project in the list are intended to be broad. Some examples of what can be measured are included. **Please note that you are expected to design an adequate analysis, which will lead to do control experiments and answer questions not explicitly stated in the list.**

Evaluation

The project will be graded up to **16 points** and the evaluation will be based on

- **Quality of the technical analysis. (7 points).** How well is the analysis executed? Are appropriate methods and techniques applied to the given research questions?
- **Discussion of results and limitations. (5 points)** How thoroughly are the results discussed in relation to the research questions? Is there a thoughtful exploration of the limitations of the approach?
- **Clarity of presentation (2 points for report + 2 oral presentation).** Is the information communicated clearly?

These evaluations will be done through the **oral presentation and the report**. Note that in science it is important not only to have a good knowledge of the topic and make an interesting analysis, but also to be able to put it in a way that can be understood by others. **You also must submit all code used to perform the analysis.**

Structure of the project

Projects will be maximum 2 pages long, excluding figures, tables and references. Please note that it is not essential to positively answer your research question, but rather to present an insightful discussion on your results, either positive or negative.

As a rule of thumb, both your presentation and report should have a structure resembling this one:

1. Introduction

- Explain the significance and interest of your research topic.
- Provide context by relating your work to topics covered in the course.

2. State your research questions

3. Methodology

- Detail the methodology, analysis, pipeline, or model used in your project.
- Provide a clear and concise overview of the steps taken in your research.
- Include any relevant technical details.

4. Results

- Present your findings in a structured manner.
- Utilize plots and visuals to enhance clarity.
- Highlight key trends, patterns, or insights from the results.

5. Discussion

- Address whether your research questions were successfully answered.
- Explain the rationale behind your findings and discuss their implications.
- Reflect on what alternative approaches could have been taken.
- Acknowledge and discuss the limitations of your chosen methodology.
- Suggest potential areas for future research based on your findings